Predicting Businesses Closure

by

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Aim

- To build a classifier that would predict whether business will close or not based on the Yelp reviews

Motivation

- To give investors a tool to better asses the risk of business failure and to help them to make an informed decision whether they should invest in given business or not

Approach to problem

- Each business in the Yelp database has several attributes/features and is labeled as ‘open’ or ‘closed’
- Binary Classification problem- we want to build a model that would decide whether a given (unlabeled) business is likely to stay open or close based on its attributes
- Data preprocessing and machine learning implemented in python 3.7/ jupyter using pandas, seaborn, scikit-learn, vaderSentiment
- Source code *https://github.com/adabebe/Yelp-Case-Study*

Yelp dataset

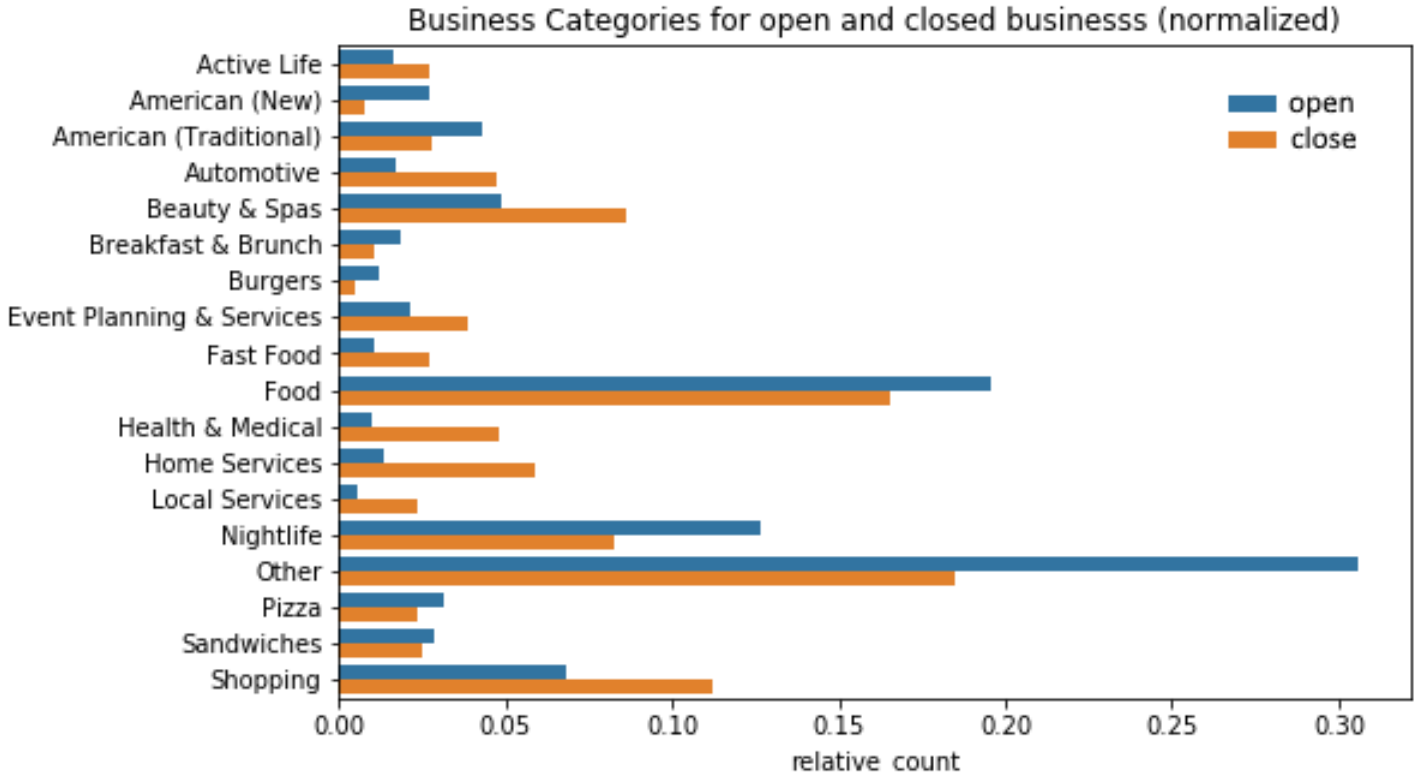
5 261 668 unique reviews
174 567 unique businesses: 27 865 closed / 146 702 open

Data-frames:

- ***Business*** **-** basic Yelp information about the business i.e. is open, star rating, number of reviews, business's location etc.
- ***Review*** - individual reviews of the businesses with star rating, text...
- ***Business Attributes*** **-** detailed information about businesses
- ***Business Hours*** - self explanatory
- ***Check-in*** - check-ins on a business
- ***Tips*** - tips written by a user on a business
- ***User*** - metadata associated with the user

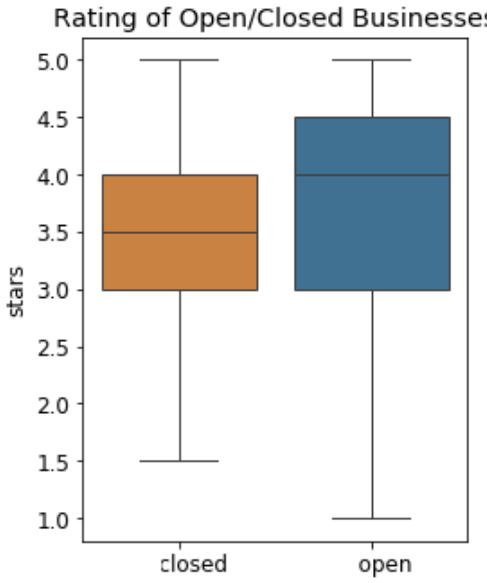
The Data: Business Categories

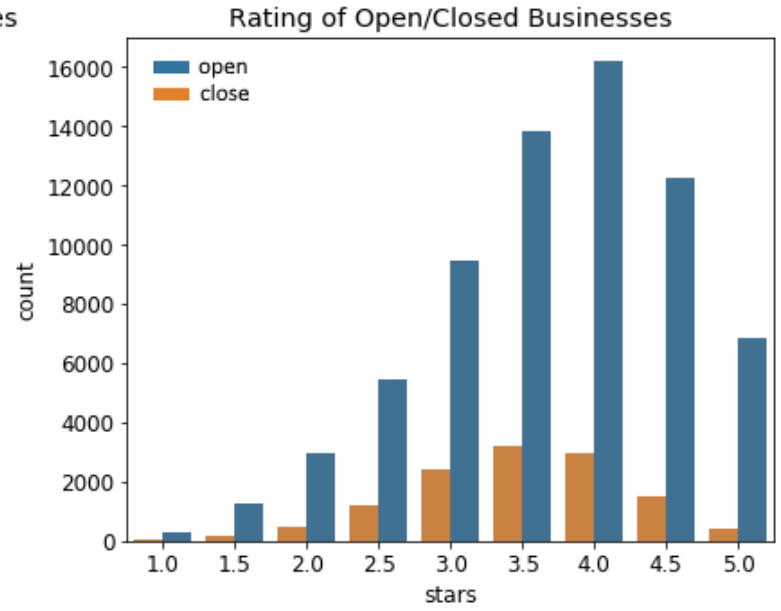
- Several business type categories separate the open and closed businesses fairly well
- E.g. There is a higher proportion of closed than open businesses in ‘health & medical’ category. On the hand, the businesses of ‘other’ category have much stronger representation in the open business group.

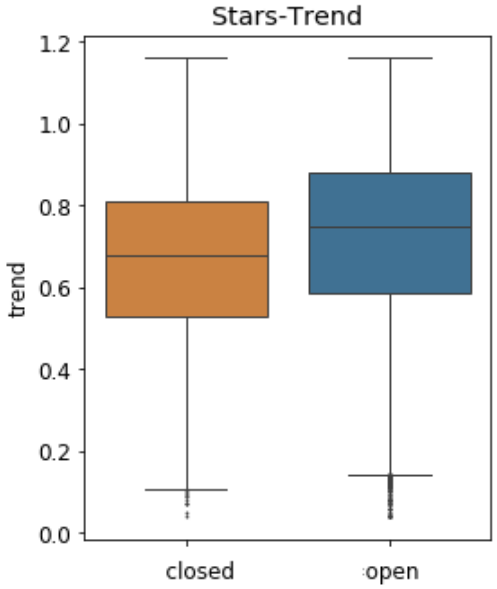


The Data: Business Star rating

- On average, open businesses have slightly higher star rating than closed ones
- We also extracted star rating over several time intervals:
- Last 180, 180 to 360, 360-540, 540 to 720 days + calculated linear temporal trend
- The temporal features have not revealed large differences between open and close businesses

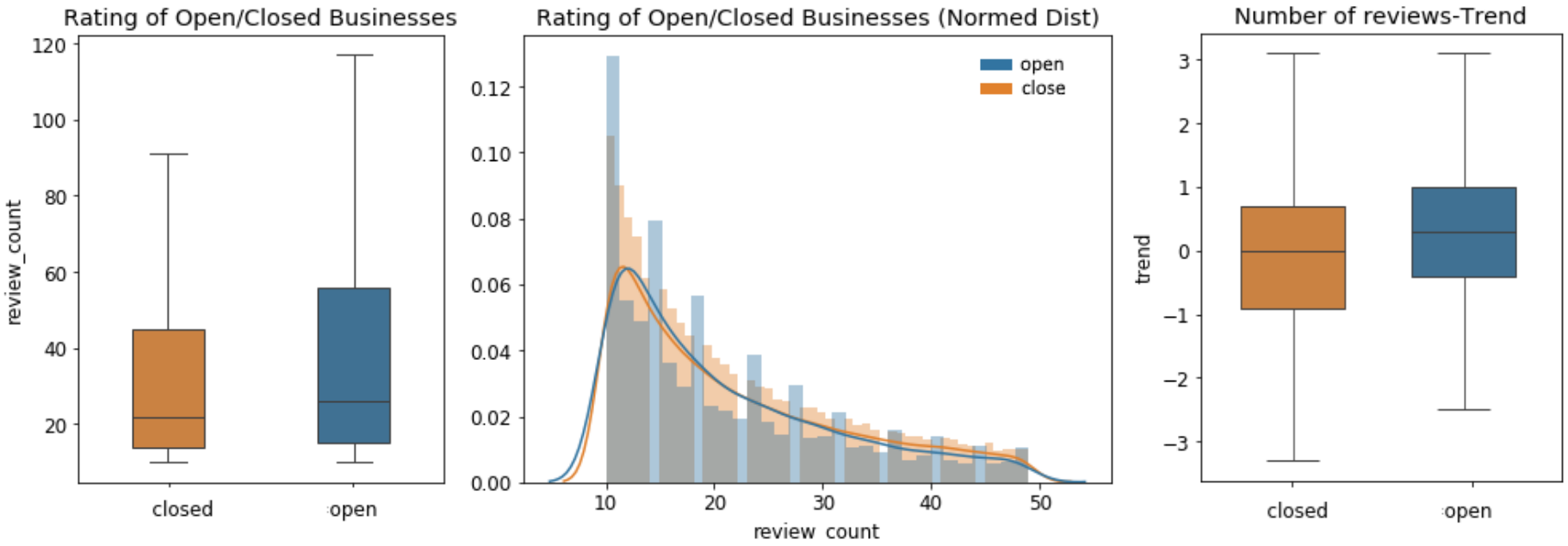






The Data: Number of reviews

- Open businesses have more reviews than closed ones
- This was also compared over several time intervals:
- Last 180, 180 to 360, 360-540, 540 to 720 days + temporal trend
- The temporal analysis revealed a drop in number of reviews in last 180 days in closed businesses



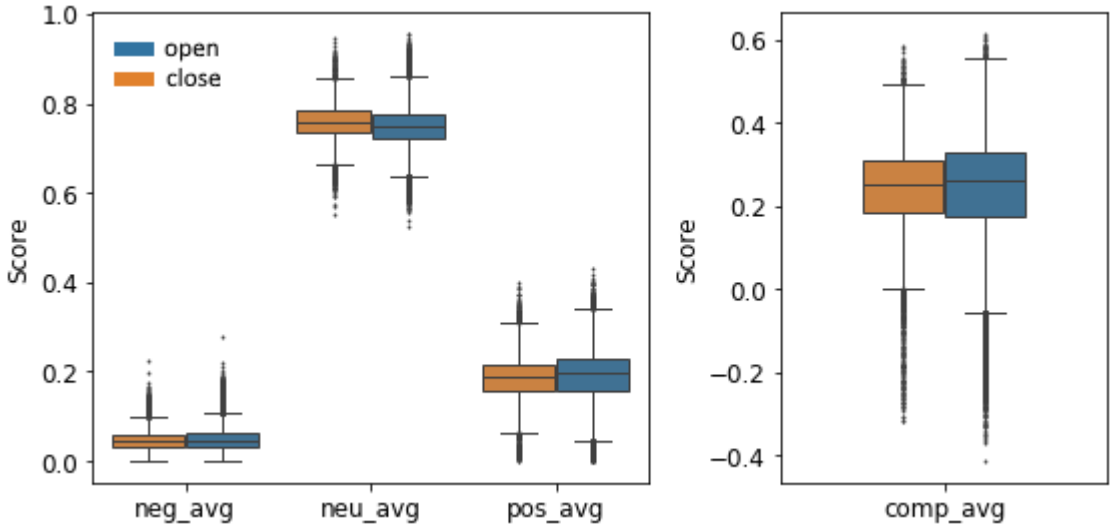
Extracting Review Sentiment

- Sentiment of each user review was analyzed using VADER tool
- Sentiment scores:
  - positive (pos) - proportions of text that is positive
  - neutral (neu) - proportions of text that is neutral
  - negative (neg) - proportions of text that is negative
  - compound (comp) - normalized composite score
- Example:
  - Disgusting food and a terrible waiter. | neg: 0.684, neu: 0.316, pos: 0.0, compound: -0.7579
  - Awesome place with strong drinks | neg: 0.0, neu: 0.288, pos: 0.712, compound: 0.8126
  - <3 <3 <3 :-* | neg: 0.0, neu: 0.208, pos: 0.792, compound: 0.886

Hutto, C.J. & Gilbert, E.E. (2014). VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text. Eighth International Conference on Weblogs and Social Media (ICWSM-14). Ann Arbor, MI, June 2014.

The Data: Review Sentiment

- There were no apparent large differences in sentiment scores between open and closed businesses
- The median compound score value is slightly higher for open businesses
- On the other hand, the open businesses seem to have also more negative scores
- This pattern was similar for measures obtained using review sentence averaging as well as using score of the sentence with the highest compound sentiment



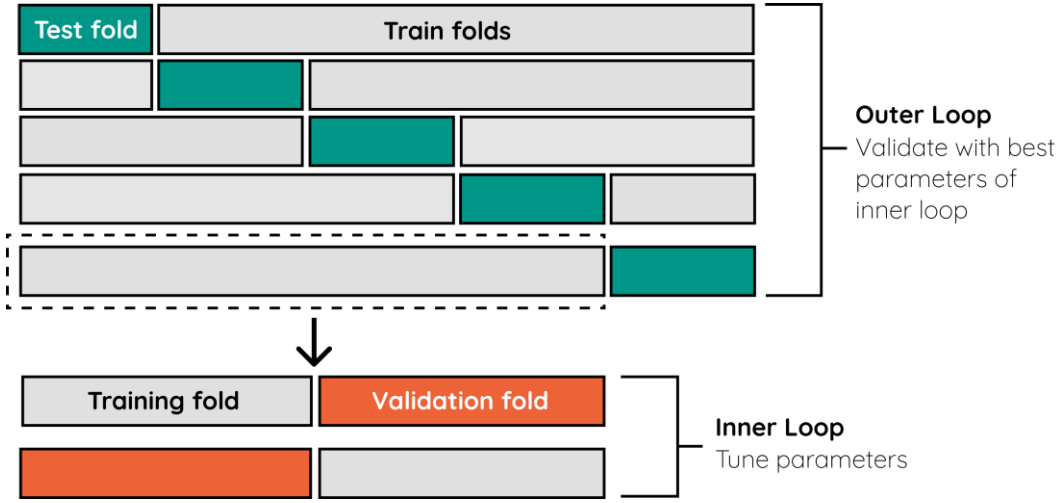
Classification feature matrix

- Extracted total 39 classification features
  - Star rating (average, last 180, 180 to 360, 360-540, 540 to 720 days, trend)
  - Review count (average, last 180, 180 to 360, 360-540, 540 to 720 days, trend)
  - Sentiment (negative, neutral, positive, compound – sentence average/ based on sentence with highest compound score)
  - Business category- One-hot encoded, resulting 18 variables

- Total of 51 906 samples (businesses) after data cleaning
  - 7 205 corresponding to ‘close’ business
  - 44 701 corresponding to ‘open’ business

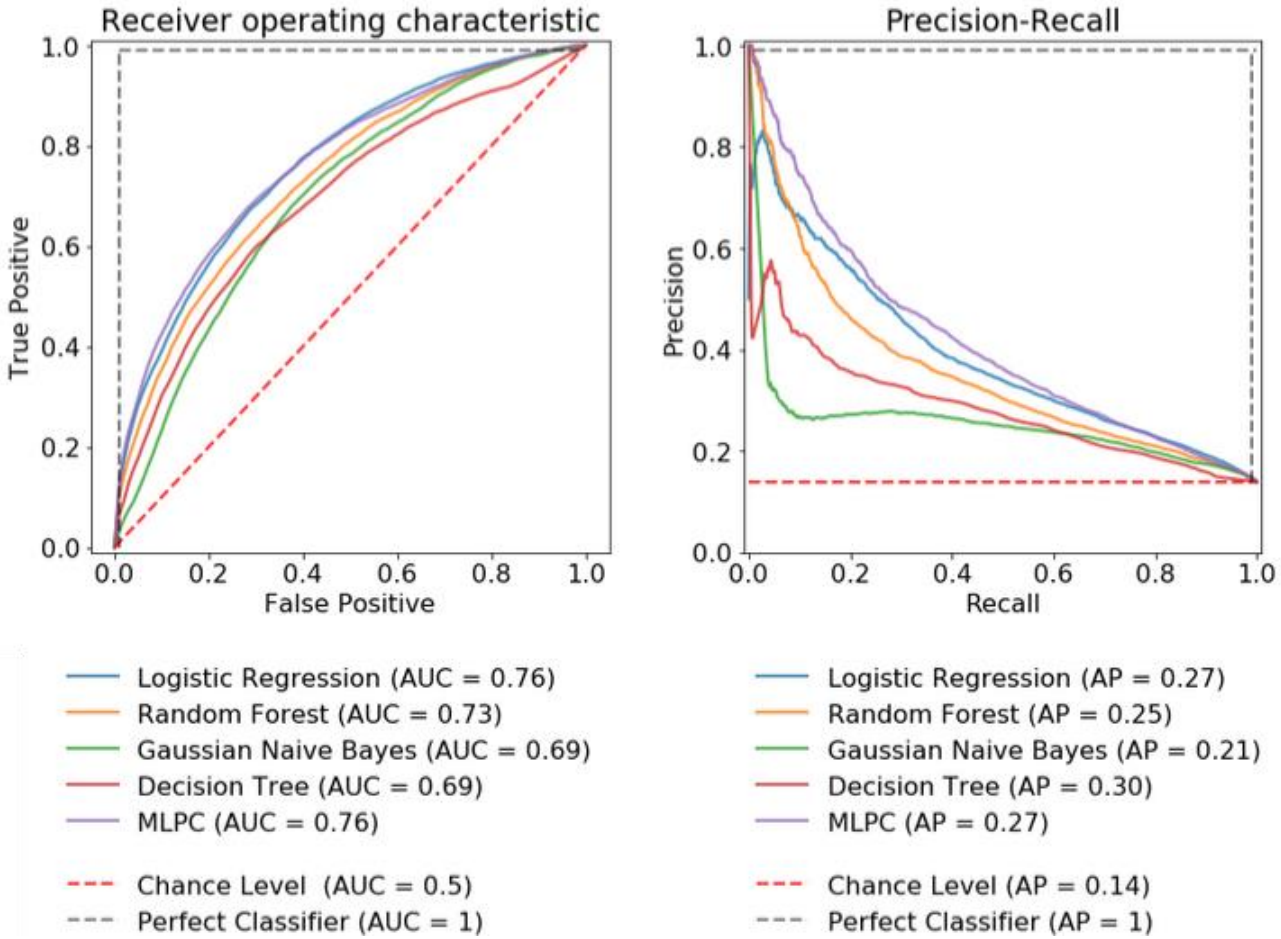
Classification Methods

- Unbalanced dataset- 86% Businesses Open / 24% Businesses Closed
- Minority class up-sampling (SMOTE) used to compensate for different ratio between classes
- ROC-AUC (Receiver Operating Characteristic- Area Under the Curve) used as a performance metric
- Nested cross-validation used to train and validate models
- Number of classification algorithms tested: Logistic regression, Random Forest, Decision Tree, Gaussian Naive Bayes, Multilayer Perceptron Neural Network



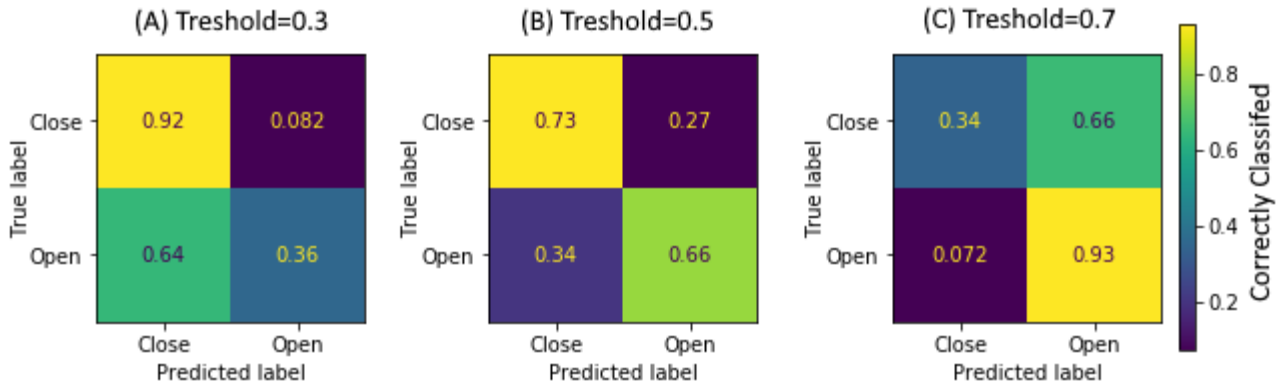
Comparison of algorithms

- All tested algorithms successfully predicted whether business will close or not (performed better than chance)
- Logistic Regression and neural network (MLPC) performed best



Logistic regression I

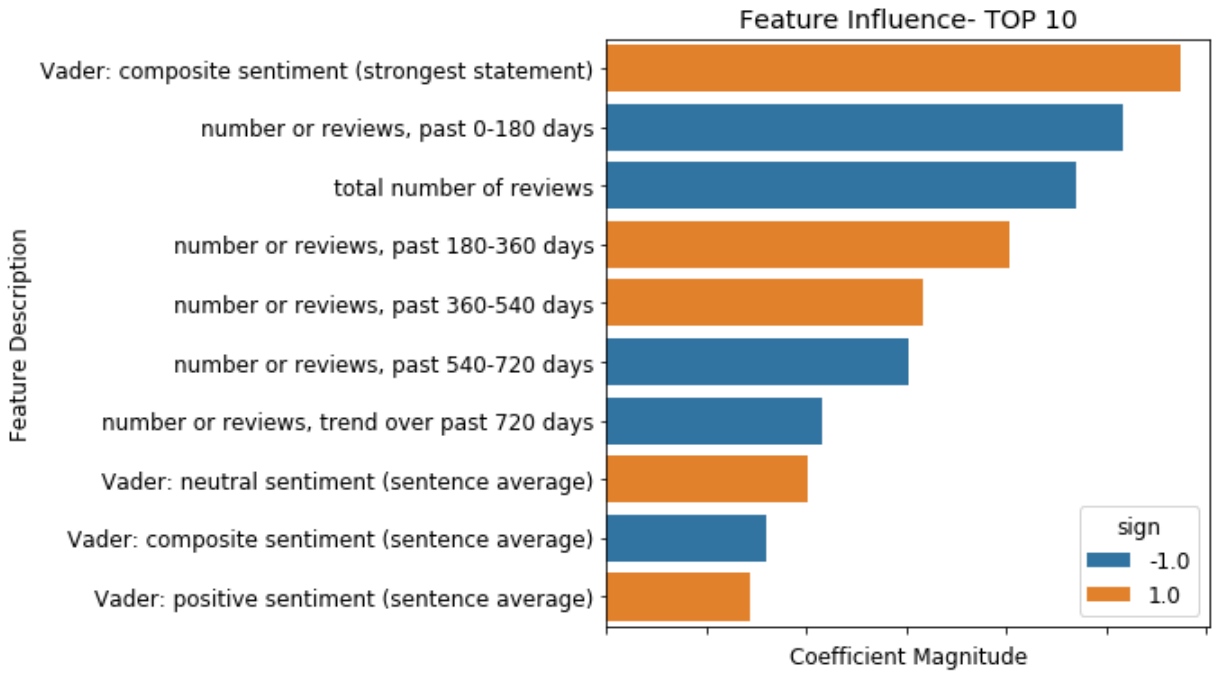
- Logistic regression amongst the best tested classification algorithms to predict whether business closes or not (AUC=0.76 / AP=0.27)
- Probabilistic output can be directly interpreted as confidence level
- Useful for solving cost-sensitive classification problems as one can modify score threshold levels to bias the decision towards one class or the other (see confusion matrices below)
- Linear model coefficients are relative interpretable (see next slide)



Logistic regression II

Feature influence estimated from logistic regression coefficients:

- Orange bars large values push the probability towards ‘CLOSE'
- Blue bars large values push the probability towards ‘OPEN'



Summary

- Here, we showed it is possible to build a machine-learning classification model that will successfully predict business closure from the Yelp review dataset.
- We have tested several different algorithms and found that all tested classification models perform better than no-skill classifier (chance) in predicting business closure based on ROC and precision-recall curves.
- The performances of all models were comparable. Nevertheless, logistic regression and artificial feed-forward neural network (Multi-layer Perceptron classifier) performed the best.
- We found that number of reviews and review sentiment were most useful for classification.
- In particular, the Vader sentiment analyses show that more useful information (beyond score and number reviews of the original Yelp dataset) can be extracted from the text content of the user reviews.

Limitations / Future Work

- Due to limited time for this project, I have restricted my analysis to 'business' and 'review' data-frames. Also, not all features of these frames were fully utilized e.g. the geolocation information was not used for classification although it might have been potentially useful.

- Classification model hyper-parameters were not fine-tuned due to time-constraints. The results might differ when fitting the models properly.

- In future work, it would be interesting to deploy automated feature engineering framework such as https://docs.featuretools.com/, which would possibly help to extract more information from the dataset.

- Given the importance of review sentiment scores in classification, it would be interesting to employ more advanced tool than VADER to analyze the review text content.